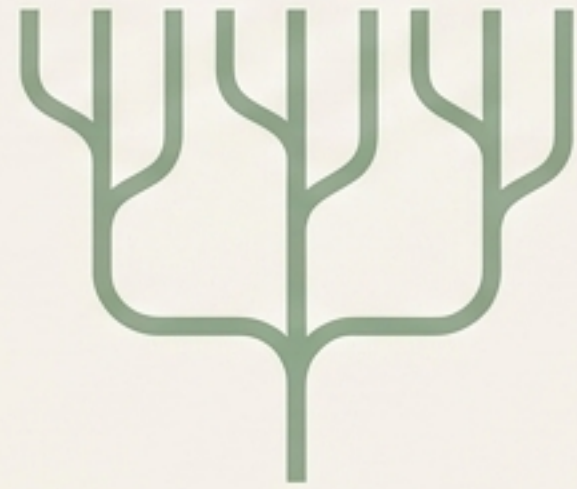


Bagging vs. Random Forest: The Ultimate Interview Question

A deep dive into the hidden mechanics
that trip up most candidates



Bagging
(Bootstrap Aggregating)



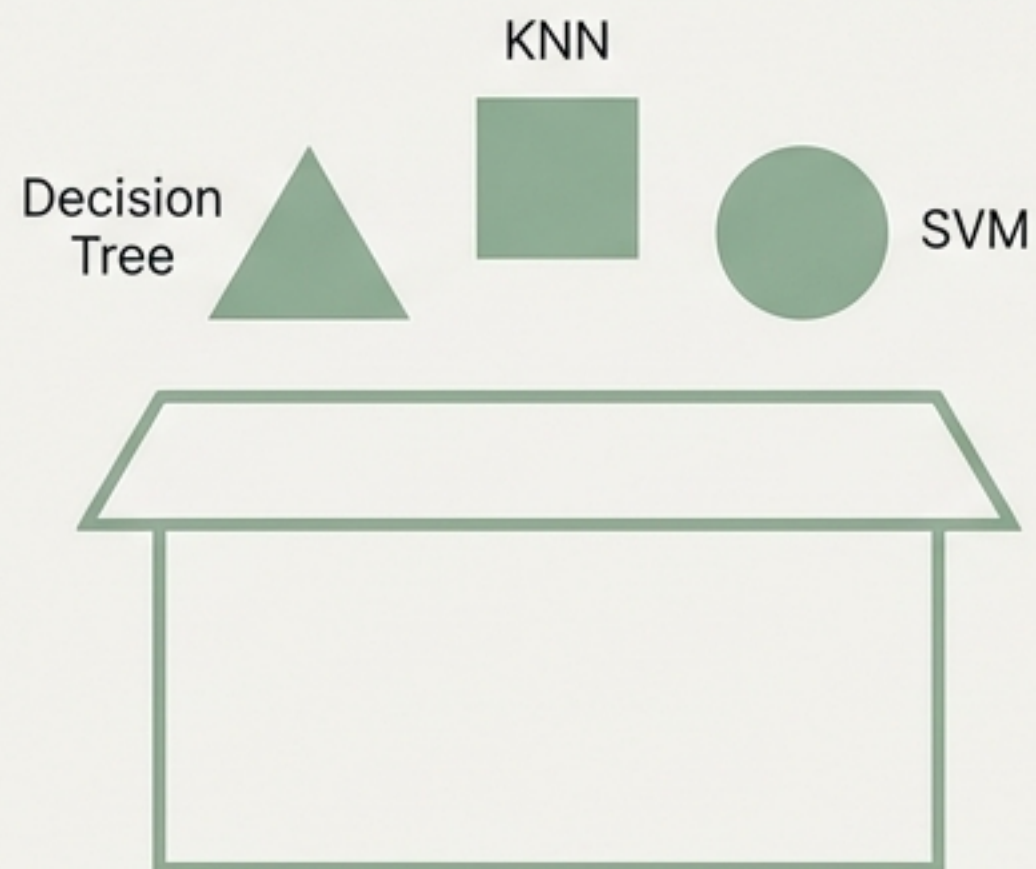
Random Forest



Most candidates stop at the base estimators.

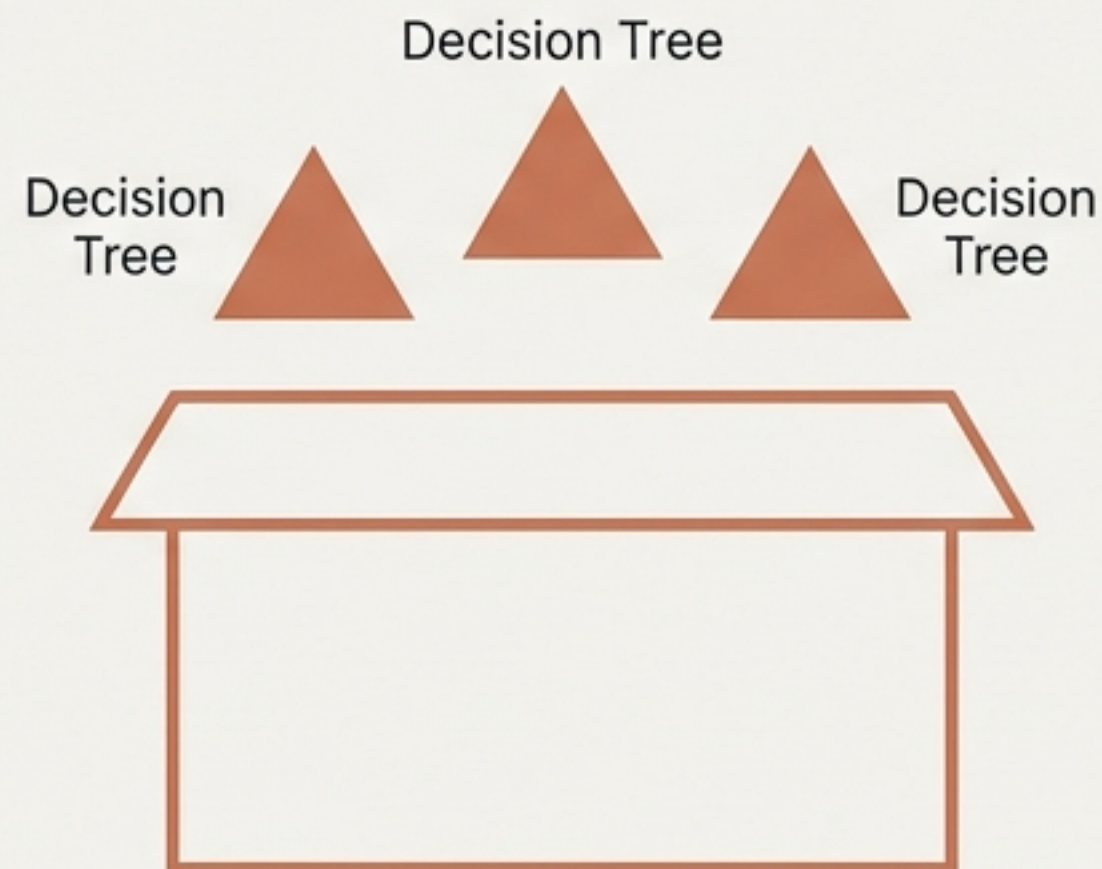
Bagging is an algorithm-agnostic technique. You can use Decision Trees, KNN, or SVMs as your base models. Random Forest is heavily restricted: its base models can only be Decision Trees.

Bagging



```
BaggingClassifier(base_estimator=KNeighborsClassifier())
```

Random Forest



```
RandomForestClassifier() // No base_estimator parameter
```


The Counter-Question.

“If I create a Bagging classifier and set all of my base estimators to Decision Trees... is it exactly a Random Forest?”

NO.

The answer is no.

Even with Decision Trees, a Bagging ensemble is not a Random Forest.

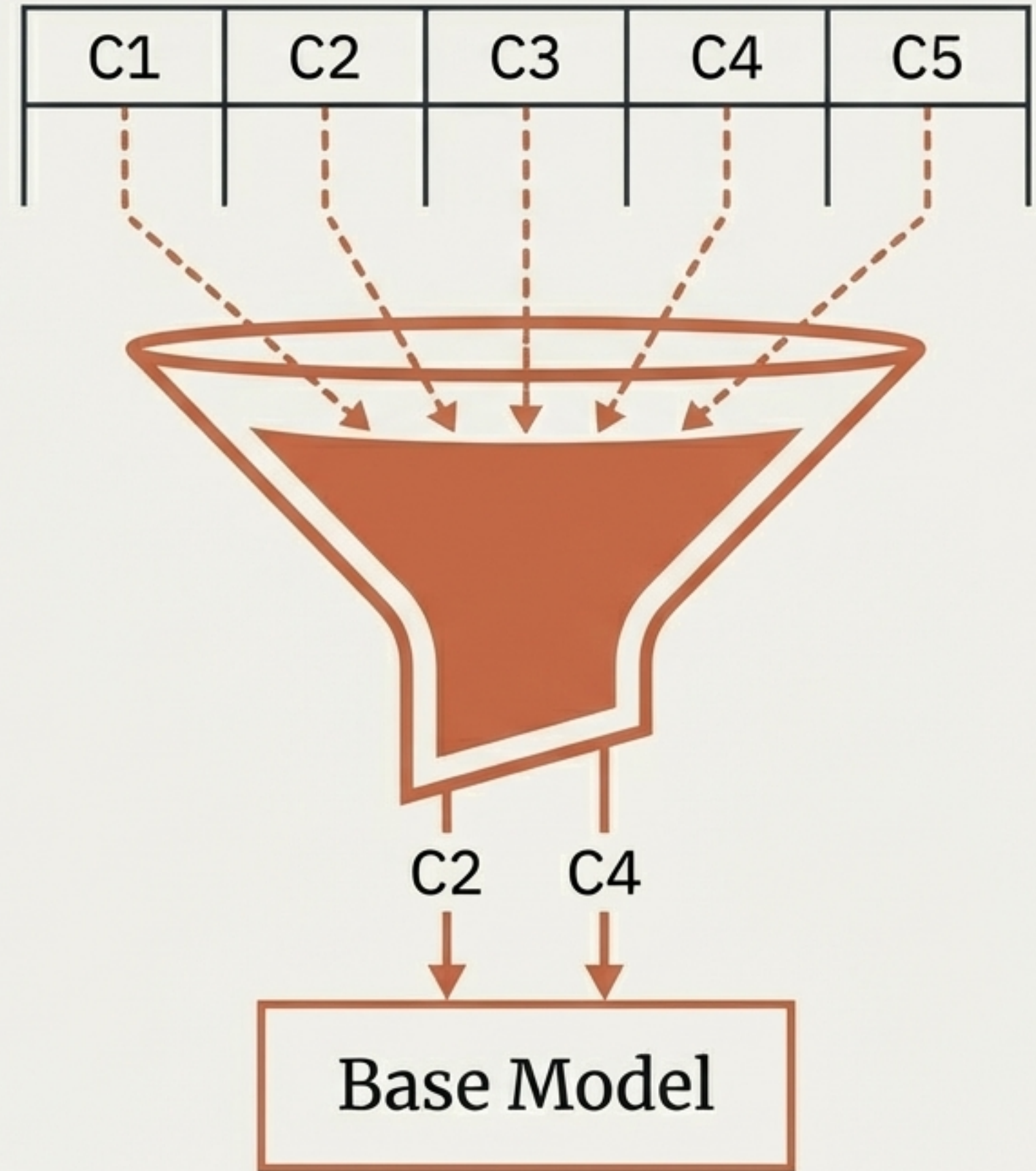
The true difference lies hidden inside how each algorithm handles Feature Selection
(Column Sampling).



A quick primer on Feature Sampling.

Instead of giving base models the entire dataset, we supply a subset using row sampling and **feature (column) sampling**.

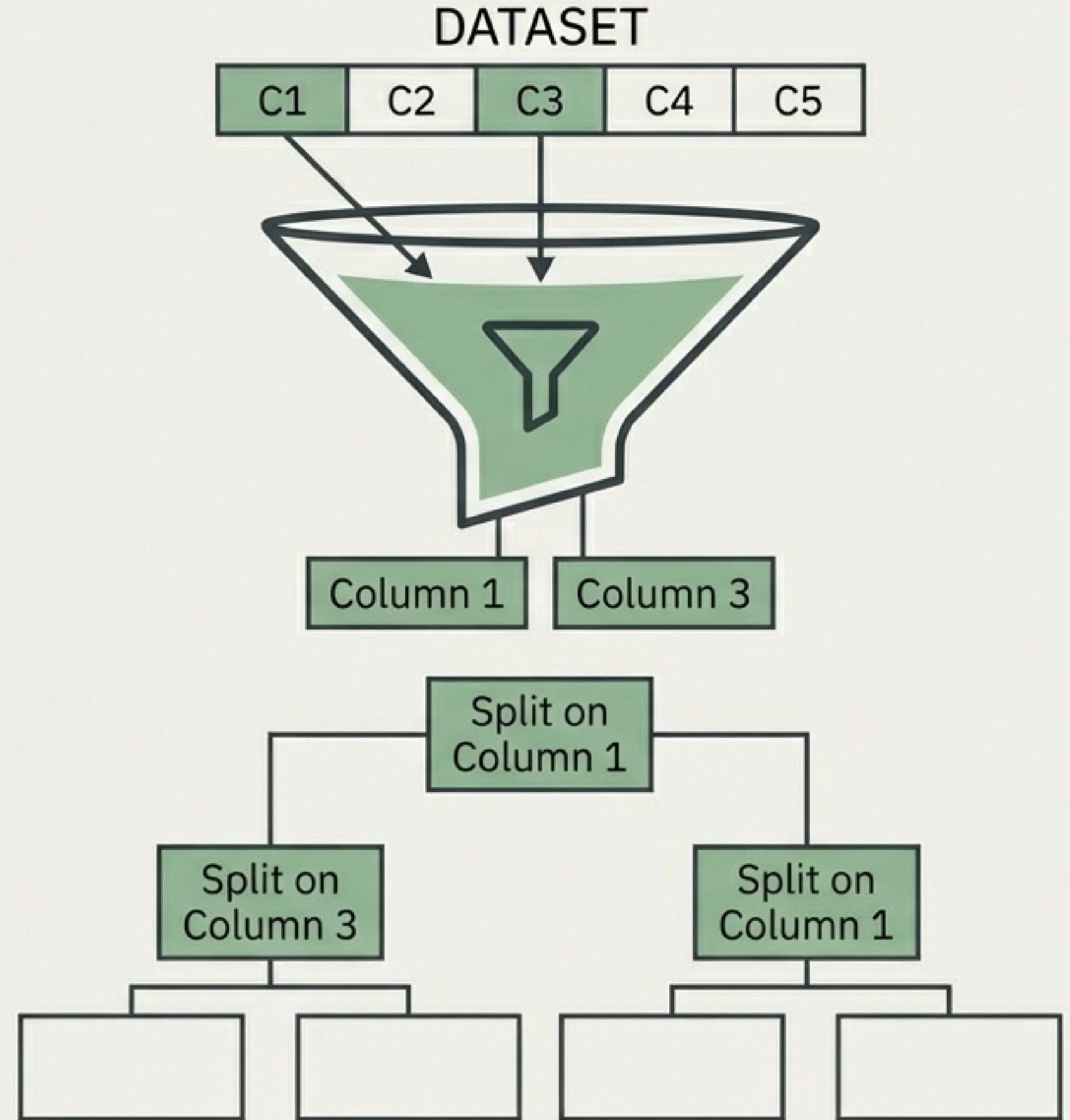
For example, selecting only 2 random columns out of 5 available.



Bagging uses Tree-Level Sampling.

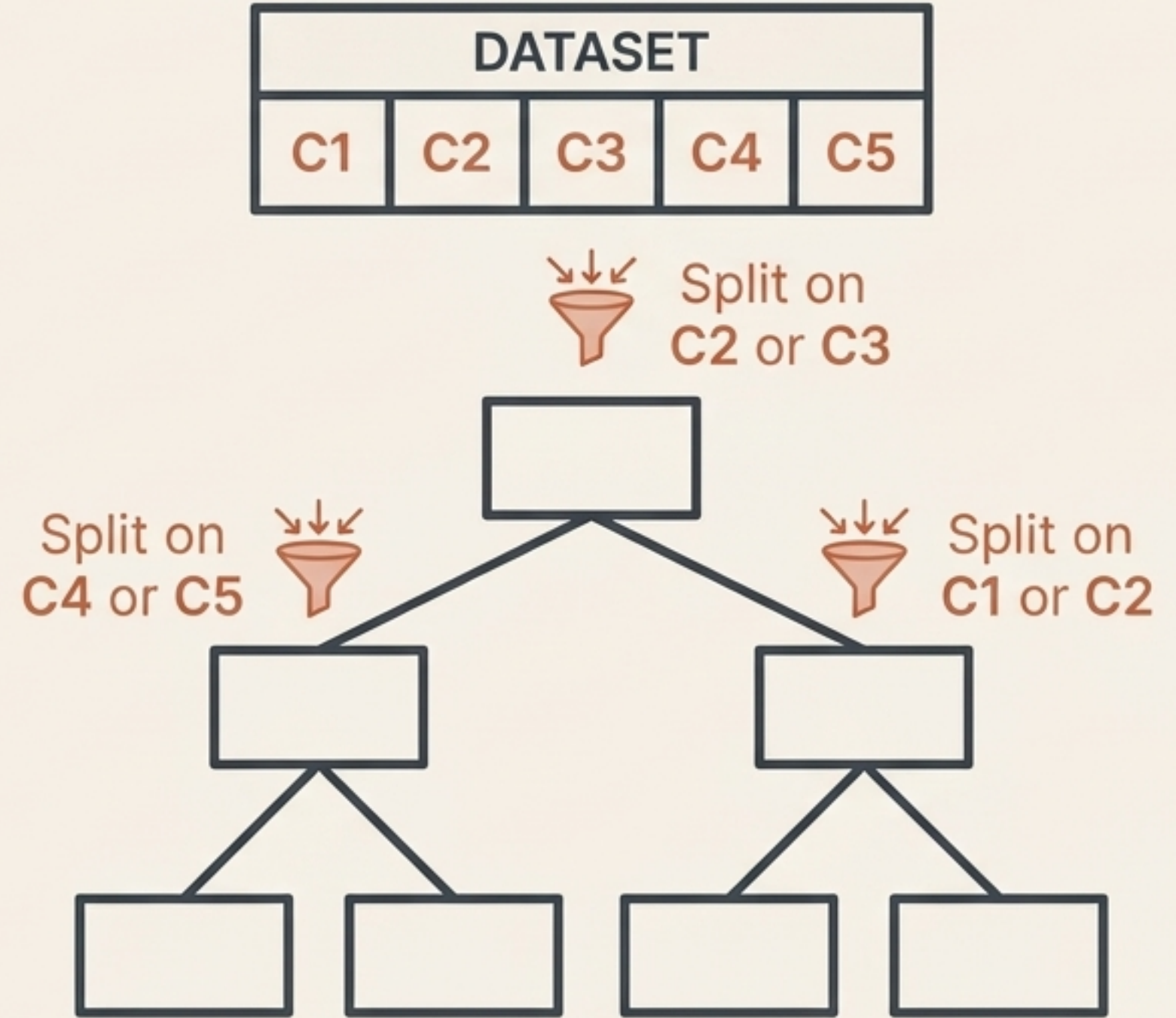
In Bagging, columns are finalized before the tree is built.

If columns 1 and 3 are **randomly selected, every single split** in that tree is restricted to only those two columns.



Random Forest uses Node-Level Sampling

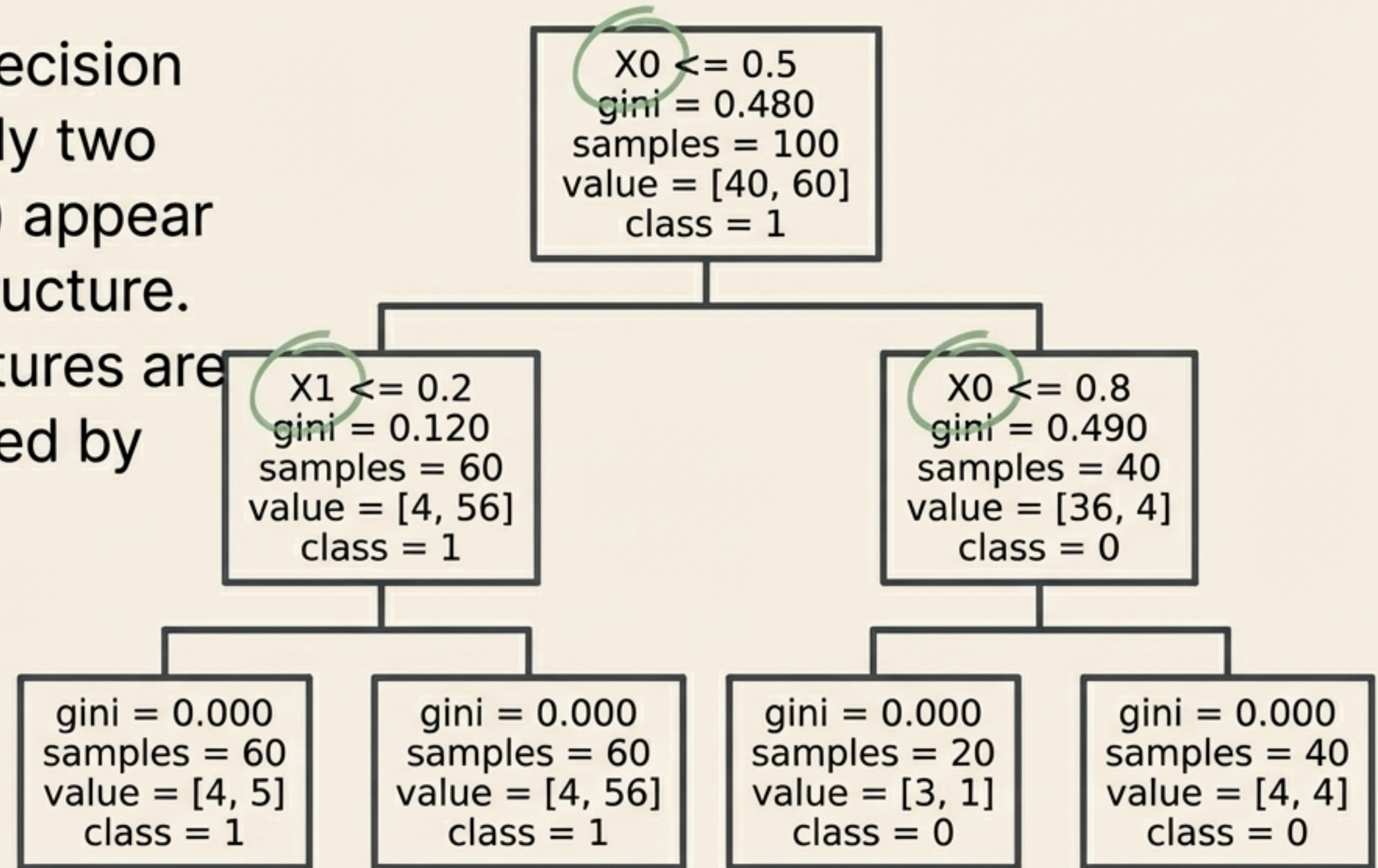
Random Forest samples columns dynamically. Every time a new node is created, it **randomly** selects a fresh subset of columns to consider for that specific split.



The Code Proof: Bagging

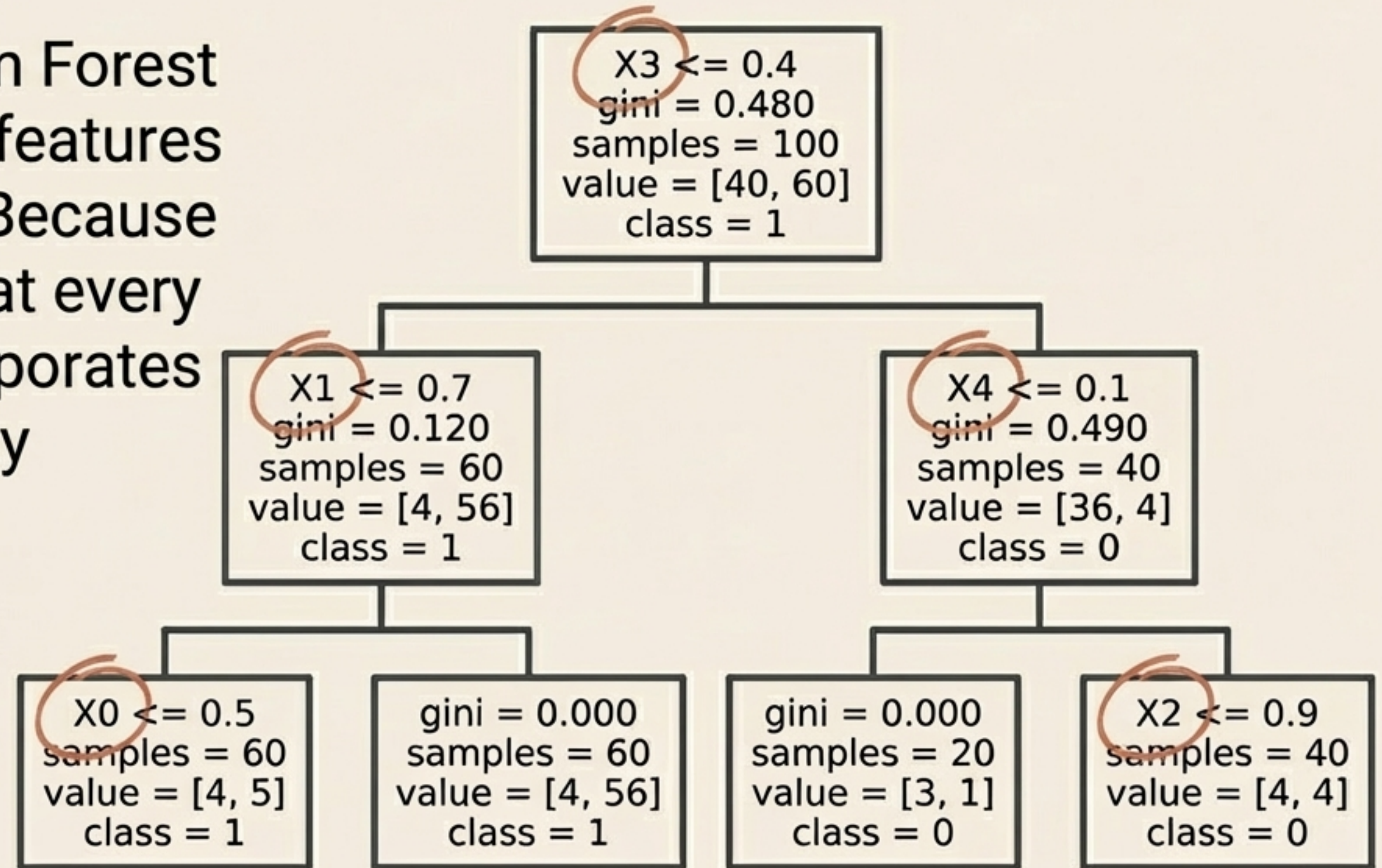
When plotting the decision tree for Bagging, only two features (X0 and X1) appear across the entire structure.

The **other three** features are completely untouched by this specific tree.



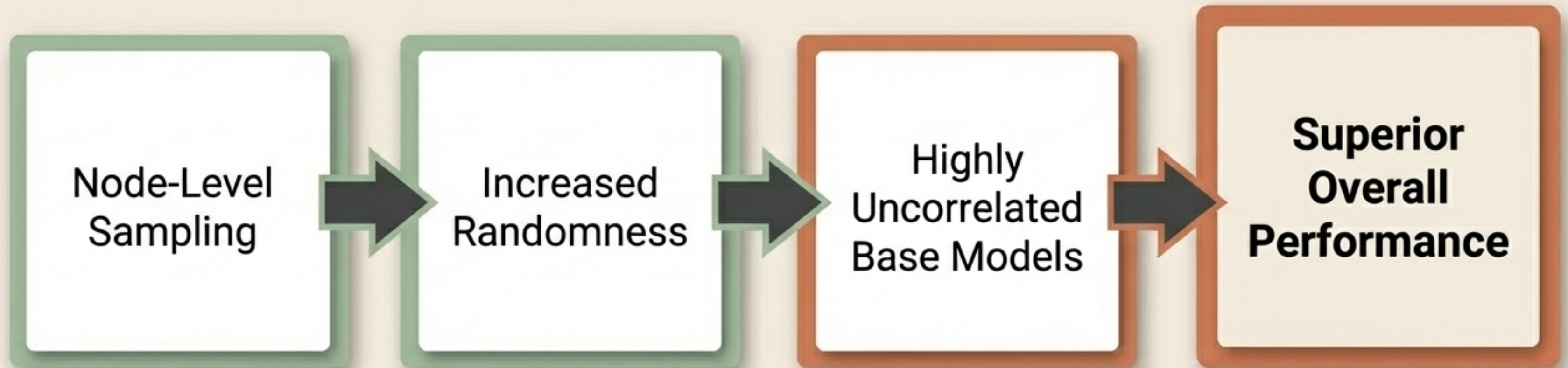
The Code Proof: Random Forest

Plotting the Random Forest reveals that all five features eventually appear. Because sampling happens at every node, the tree incorporates a much wider variety of data.



Why Node-Level Sampling wins.

Node-level sampling forces maximum randomness into the ensemble. The greater the differences between the base models, the stronger the final prediction. This is why Random Forest typically outperforms Bagging.



The Interview Cheat Sheet.

Save this for your next technical interview.

	Bagging	Random Forest
Base Estimators	Any	Decision Trees Only
Scikit-Learn Parameter	Uses <code>base_estimator</code>	Hardcoded
Feature Sampling Level	Tree-Level	Node-Level
Randomness & Performance	Moderate	Maximum